Explainable AI Background and Literature Review

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November 2019

1 Abstract

1.1 Purpose Of This Document

This review is intended to document the research that has been undertaken so far by the author during the first 3 months of their PhD, giving clarity and context to the current focus of the project and providing substantial documentation to be recycled later in the PhD timeline. After review, the supervisor and author will agree a suitable line of enquiry to pursue during the next stage of the PhD. Similarly, this document is intended to raise any concerns the supervisor may have at this point.

1.2 Overview of document

This document begins with a high level overview of explainable artificial intelligence (AI) motivating its study and providing an indication of its current state in academia, industry and government. Study of explainable AI has so far been focused on developing explainable models, that is, ways of explaining to a user a decision that has been made by an AI model [1]. We continue with discussion of state-of-the-art in explainable machine learning.

Next, we argue the importance of considering historical inter-disciplinary theory when developing systems that are designed to explain their behaviour. We then explore philosophical, psychological and sociological background in explanations and how this may relate to AI. We then introduce multi-agents systems and their prevalence in modern technology. We discuss the growing importance of considering explanations for these mechanisms. We conclude by combining this theoretical perspective with technical state-of-the-art in the area of explanations within multi-agent systems.

2 Introduction

It's Friday afternoon and you're eagerly anticipating 5pm, you're flying to Spain tomorrow for 2 weeks of sun, wine and no emails. As you're getting the Tube home, you catch a glimpse of the Metro headline:

NEW CREDIT SYSTEM REVOLUTIONISES THE WAY WE PAY: NO MORE QUEUES!

You're skeptical but would definitely appreciate a shorter queue for a coffee in the morning. You don't care anyway, you're off on holiday tomorrow and can already feel the sand between your toes. Saturday morning arrives and you're on the way to the airport, when you get there you notice the new automated doors. They look like passport control gates. "Strange, that must be the new credit system, shouldn't be a problem I've always paid my taxes" you reassure yourself. The gates turn green and open as the lady in front of you passes through without a problem. As the laser scans over your face you appreciate the magic of technology, "No more fraud, no more criminals on the loose, no more queues, this is fantastic!" But the light turns red and you are denied access. "What, why?" You're confused and angry as it becomes increasingly clear you will not be able to get on your flight, you will not be going on holiday. "I demand answers" you shout angrily at the door but your only response is a filthy look from the person behind you and a flashing red light...

This story sounds dystopian, like the opening to a 1970s science fiction novel, yet it bares similarities to China's ongoing experiment in social credit. Other recent events including the fatal driverless car crash in Arizona in 2018 [2] and the racial bias embedded in a US recidivism algorithm exposed by ProPublica [3] illuminate concerning behaviour of high stakes technology, evidence of how artificial intelligence (AI) become increasingly entrenched in society. Further, recent advances in computational power has meant technology is capable of great performance (sometimes better than humans) on increasingly complex tasks. Tension arises as technology systems are integrated into traditionally human systems, forming a hybrid decision-making process that blur the boundaries of responsibility and accountability.

Technology has infiltrated human life at an unprecedented rate, now more people have access to a mobile phone than they do to clean water. Innovation is often encouraged without consideration of its long term impact yet technology has the capacity to cause serious problems for humanity in the future: widespread unemployment, breakdown of social skills or some dystopian future like the story above. We argue that we must start thinking about whether *we should* alongside whether *we could*. To encourage this new perspective on innovation there needs to be in a systemic shift towards responsible AI. In this domain, we encourage honest AI as technology that is aware of its weaknesses, bias and underlying behaviour. This project focuses on AI that can explain itself as one

tool towards honesty.

The interest in explainable artificial intelligence is proportional with the rise in use of black box models over the past decade. Black box models are systems whose behaviour is unintelligible by a human. Prior to the 21st century artificial intelligence was predominately non data focused, instead concerned with hard coding human expertise into systems. However, with the advent of the GPU alongside the increase in availability of data artificial intelligence changed to be predominately based on learning from data. As the volume of data increased, so did the complexity of algorithms. They can now leverage billions of parameters to make highly accurate decisions yet their behaviour is now outside the remit of human understanding. Explainable AI began as an endeavour to understand the behaviour of these models and this use case is what most people consider explainable AI. However, this project argues that there is more to explainable AI than explaining deep learning which we discuss further below.

2.1 Explanations For Trust

Lee et al. argue that humans respond to technology socially, resulting in misguided reliance on automated systems [4]. Trust has existed for a long time as a way to define people's relationships and has recently been applied to technological relationships [4]. Lee et al. explore the trust relationship between humans and automation, drawing on the results from Mosier et al. to discuss how various factors including low self confidence and perceived reliability can result in "automation bias" [5].

Richardson et al. discuss the reasons we need explainability of AI systems [6], giving 3 levels of demand: not helpful, beneficial, necessary. They argue there are situations where an explanation is necessary (intelligent tutoring systems), or beneficial (a medical diagnostic black box). However, they argue there exist situations where an explanation is not beneficial(a recommender system that gives advice based on a well-known algorithm). While we agree that there exists a scale to measure the effectiveness of an explanation, we argue that this issue is far more complex. For example, while an explanation is not needed for the legal accountability of a reccommmender system, it may allow the user to feel more comfortable using the technology.

Jaume et al. argue that technology is affecting people architectonically, impacting society as whole [7]. The advent of the Internet allowed technology to infiltrate society without a general understanding of underlying mechanisms or consequences. Bridle describes the Internet as "a global system of great power and energy that nevertheless retains the aura of something numinous, almost impossible to grasp" [8]. The 2018 dotEveryone report citemillerDot documents how, 57 years after the mobile phone, 30 years after the invention of the world wide web and 16 years after the invention of Facebook there exists a universal lack of understanding of technology. Katz et al. have discussed how this tension

between society and technology has led to accelerated globalisation creating a widening skills gap [9]. This, alongside the "The impulse to humanize algorithms" [10], make it impossible for the general public to fully understand the ethical challenges brought about by emerging technologies [10]. Salmon et al. argue that the problems posed by algorithms individually are exacerbated when they interact under unexpected behaviours, with the capacity to completely overwhelm the system it was designed to augment [11].

The Cambridge Analytica scandal [12] brought into question the behaviour of big technology companies who rely on the digital gap to assert their authority and make decisions on behalf of society: " They've ducked the easy questions for so long that it's reasonable to suspect that they doubt we will like the answers. " [13]. In response, users now want to know why their technology works in certain ways.

Government has responded to this societal shift with the data protection law, comprised of the GDPR and DPA, which works to regulate the use of personal data. Alongside this law, the government is working on further standardisation and guidelines to regulate AI.

In this space, the Independent Comissioner's Office (ICO) has recently published guidelines on explaining AI [14]. Whilst a step in the right direction, literature like this has the potential to cause further problems. One of these problems is the lack of standardised approach across industry, academia and government. Interdisciplinary researchers all agree that explainable AI is a big issue but there is no standardised direction of work [15]. This is exemplified by looking further at the ICO explainable AI guidance that discusses in length the sort of training that should be provided to the explanation "implementer". It appears that this document considers explainable AI from the perspective of auditability. The lack of standardised approach highlights two key issues:

- Lack of communication between academia and industry, "very few stop to check what they are agreeing to, and to find out what explainable AI means to other people involved in the discussion" [15]
- Lack of clarity in definitions

Some further risks of explainable AI are discussed below.

2.2 Risks of explainable AI

In their report, the ICO discuss some of the dangers to the industry of explainable AI [14]. They argue that increased distrust, exposure of commercially sensitive or private material and danger of malicious users gaming the system all present risks that must be mitigated in future design of explainable AI. Whilst the ICO considers mitigation from an organizational perspective, Molnar et al.

argue these risks should be considered from a technical perspective [16]. Similarly, Kascper et al. consider the security risks associated with counterfactual explanations [17].

2.3 What is meant by an Explanation

Explainable AI research struggles with the lack of standardised definitions and the gap between approaches is most significant between machine learning and the explanation sciences (including philosophy, cognitive science and sociology) [15]. Mittelstadt et al. argue that machine learning re-purposed the definition of explanation, ignoring centuries of research in the explanation sciences.

In the literature, the terms, explainability, interpretability, explanations, transparency, trust, justification, rationalisation can all be defined differently having a dramatic effect on the direction of the corresponding research.

Lipton presents his definitions for these related terms, arguing that interpretability is ill defined within the literature but is composed of several ideas including transparency and post-hoc interpretability. Lipton's paper is clearly from a machine learning perspective, in contrast to the ICO explainability guidelines [14], whose definition of explanation includes those that explain software design rationale, data collection methods and model explanations.

Doran et al. examine a corpus of literature from various well established communities in explainable AI including the Conference on Neural Information Processing Systems (NIPS). Here, Doran et al. define differences between opaque systems that offer no insight to the algorithmic workings, comprehensible systems that provide symbolic and user friendly explanations of algorithmic workings and interpretable systems that can be mathematically analysed to understand the underlying algorithmic workings [18]. Under these definitions of interpretability and comprehensability, Doran et al. argue that both approaches lack a way of providing a comprehensive line of reasoning to the user, something that they argue, is essential for formulating "truly explainable AI" [18].

2.4 Philosophical Background of Explanations

We have discussed how many modern approaches to explainable AI neglect the vast theoretical research within the explanation sciences. This section therefore examines the philosophical background of explanation science, exploring the complex relationship between causality and explanation and develops a line of thought over what constitutes a *good* explanation.

Throughout history, explanations have been viewed as deductive arguments, statistical relations, schema applications, analogical comparisons, causal relations, and linguistic acts [19]. Thagard argues that despite the value of these

practices in evaluating explanations, causality is at the core of all explanations [19].

2.4.1 Causality and Counterfactuals

Aristotle believed that causation and explanation are intrinsically linked, defining exactly four causes: formal, material, efficient and final [20]. Recent debate over Aristotle's four causes argues they should be renamed "four becauses". Ultimately, Aristotle's philosophy of causation states that when asking "Why something is" we seek the "primary source of its being" [21]. Over the consequent centuries philosophers such as Thomas Aquinas embedded Aristotle's idea of causation in religion to encompass the idea that God created everything from nothing [22]. With the birth of modern science, philosophers such as Hobbes and Descartes rejected ideas of formal causation and expressed causation in terms of physical movement. The theories of this time were a product of science and religion [22].

Hume refutes causal necessity under the basis there is no way of rationally justifying that given a cause, the effect must follow [23]. He argues that causes cannot be known a-priori and instead gives two definitions of causality that capture the idea that for a single event we cannot possibly know what caused it, however, after observing many conjunctive chains of cause and effect whilst necessary cause, it gives the feeling that it is. These definitions first capture the idea of regular succession and then attach a psychological experience to the succession [24]. Mackie, like many philosophers of the time, criticises Hume's conjoining of psychology and philosophy. However, Mackie praises Hume for raising questions over what constitutes necessary connection [24].

While Hume's view of necessary relationships and definitions of causes are widely accepted today, the subtlety of Hume's arguments has been debated and expanded by multiple philosophers. Lewis argues that Hume's first definition is susceptible to epiphenomena [25] and that Hume's second definition expresses a counterfactual view of causation. Similarly, Maackie argues that the item for which we must seek a pscyhological reaction is not the Humean idea of an a-priori link between similar events but a link that is expressed by counterfactual conditions [24].

Woodward argues that perception of unspecificity has caused philosophers to overlook the counterfactual view of causation [26]. Miller argues that competing definitions to the counterfactual view of causality, interventionalist theories; probabilistic theories; transference theories, are all themselves based on counterfactual reasoning [27]. Lipe argues that while counterfactual reasoning is preferable it is often impossible due to a lack of available information [28]. Lipe et al. argue that in the place of counterfactual data, it is possible to use covariational data or alternative explanations as substitutes.

2.4.2 Attribution Theory

When studying causal reasoning it is important to also consider the study of how humans make causal explanations (attribution theory). Miller claims that attribution theory is integral to the future study of explainable AI [27]. Lipe at al. [28] argue that counterfactual reasoning underpins contemporary studies in attribution theory [29].

Kelley argues there are two cases in attribution theory:

- Where the attributor has information from multiple events
- Where the attributor has information from a single event

Kelley argues that the first case requires allows an event to be attributed to one of its causes by studying covariance over time. The second case requires the application of configuration concepts where the effect of a first cause can be ignored if other causes are also present. Kelley questions the process of "taking causes into account" and introduces two types of schema for causes, multiple necessary causes that depict a set of events all necessary for an effect and multiple sufficient causes which show there are multiple ways to cause a particular event. [29].

Miller argues that causal chains play an important part in connecting causality and explanations. Humans cognitively select the relevant part of a causal chain to generate an explanation [27]. In explainable AI, difficulty arises when determining relevant segments of a causal chain, [30] argues that humans prefer to use voluntary actions within causal sequences to explain and [31] discusses the different types of causal chain and how these can be linked to different types of explanations [31]. Mittelstadt et al. argue that the extent to which causal chains are extracted constitute different types of explanation [15] expressed as "everyday" or "scientific" [27]. Miller argues that everyday explanations are those that are required by users of AI technology [27].

2.4.3 Contrastive Explanations

Lombrozo questions the relationship between explanations and causal reasoning, challenging theories that neglect the impact of prior knowledge on explanation based reasoning [32]. Non-causal explanations have been studied in mathematics and introduced in science [33] but many philosophers contest the existence of non-causal explanations [34].

Miller argues that the most important finding from the explanation sciences is that explanations are inherently contrastive [27], he argues that the size of the causal chain is less important than the fact the human brain is wired to understand contrastive explanations. A contrastive explanation differs from a counterfactual case in the sense that contrastive explanations consider hypothetical outcomes whereas counterfactual explanations consider hypothetical causes

[27]. Lipton argues that causal explanations are able to explain without being explainable themselves. Therefore, all "why" questions need contrastive explanations [35]. Miller argues that most work in this area considers contrastive questions instead of contrastive explanations [27]. Martens and Provost define a framework for providing contrastive explanations [36]. McGill et al. conduct user studies to evaluate the difference between counterfactual and contrastive explanations, reporting a tendency to engage with contrastive thinking [37].

3 Explainable Machine Learning

The section on interdisciplinary approaches to explainability discusses the theoretical background underpinning explanation science where we establish that human explanations are a complex phenomena dependent on many intrinsic and extrinsic factors. Tim Miller argues that an explanation consists of a cognitive causal attribution, some generation of the explandum product and a social process that conveys the explanation [27]. To contextualise these components consider explaining why you slept through your alarm this morning. First, you may look back to the previous evening collecting possible causes such as, a long workout at the gym after work, an extra cup of tea before bed and staying up for an extra hour to watch some TV. Second, you may start to select causes to form the most likely, convincing or - if presenting an explanation to your boss - the most appropriate. Thirdly, you decide om a method of delivery of the explanation, do you lie by omission, write an apology letter or turn the whole thing into a joke?. We argue that all these processes are equally important when generating explanations - your boss would not be interested in an explanation that listed all the possible causes you could come up with, your co-workers equally so with an apologetic letter. We argue that these processes should be considered equally when generating explanations for artificial intelligence. Anjomshoae et al. define the three explanation phases as explanation generation, explanation communication and explanation review and use these definitions to provide a road-map for future study [38]. Similarly, Neering et al. suggest that most approaches to Explainable AI focus on well delineated problems that can be situated within one of the three phases of an explanation [39]. They propose a model that moves beyond this monolithic structure to address all three phases of explaining AI. We build on this theory to define our 5 stages of explainable AI that can be used to categorise recent approaches and highlight future work. Crucially we argue that future work should focus on the communication between each stage to result in interactive, adaptable explanations. Our 5 stages are detailed below.

1. **Stage 1: Finding Possible Causes**

   The introduction discusses the complicated philosophical background underpinning the complex relationship between explanation and causality. We argue that there is an undeniable link between the two, in agreement with Miller, who argues that an explanation is some segment from the causal chain for a particular event. Stage 1 attempts to uncover this causal segment for a particular event. Broadly speaking, this stage collects possible reasons for a particular outcome. When applied to explaining AI this could mean performing some form of feature importance, analysing the structure of a decision tree or citing goals (of a Belief Desire Intention agent).

2. **Stage 2: Developing an Explanation**

   The function of Stage 2 is more nuanced than Stage 1 and is often over-

looked by computer science research in the explainable domain. It asks the question, how do we go from a very large collection of possible causes to the best possible explanation? Deciding what constitutes the **best** explanation is extremely difficult with many approaches making their own argument. When applied to explainable AI, work in Stage 2 should be interdisciplinary in nature and includes work that generates contrastive/counterfactual explanations, context-aware explanations and relevance estimations for explanations.

3. **Stage 3: Communicating the Explanation**

   This stage is concerned with both the presentation and the social process of an explanation. The presentation of the explanation can include methods such as textual explanations, graphical explanations and argument based explanations. The social process of an explanation is more nuanced and again more frequently overlooked by computer science output but more recently, has been garnering more attention. This process can be instantiated through methods such as interactive explanations, learning explanations, explanation via chat-bot. Most work in this phase is concerned with techniques that improve the human interpretability of the explanation. However, explanation communication must also be considered between multiple agents or robots. These types of explanation is discussed further in a later section.

4. **Stage 4: Evaluating the Explanation**

   The work in this stage attempts to address one of the main barriers currently facing the domain of explainable AI - what determines the success of explainable AI? Methods in this stage may consider the robustness, fairness and security of explanations as well as efficiency metrics or user satisfaction.

5. **Stage 5: Deploying the Explanation**

   This stage is intended to bridge the gap between the demand in industry for explainable AI and the predominately laboratory based research covered by the previous four phases when considered individually. Work in this space must devise systems that communicate between the four phases defined above as well as apply the theory to a real-world problem.

In the Introduction we identify a significant issue within explainable AI as being the lack of standardised definitions. This issue is reflected again is the variable and often confusing approaches to review the literature in this space. Molnar uses a taxonomy of four categories, post-hoc or intrinsic; result of the method; model specific model agnostic and local or global methods [16]. Other attempts at classification include Gilpin et al. [40] who recognise the difficulty in categorising the entire space of explainable AI and focus their effort instead on categorising explainable methods in deep neural architectures. Zhang et al. approach classification from a specific domain, surveying methods surrounding

explainable reccommender systems [41]. These approaches focus on inherent properties of the explanation method to form classification. In contrast, Doshi-Valez and Kim change their perspective through their taxonomy, classifying existing literature using evaluation approaches [42]. Adadi et al. attempt to synthesise these approaches by creating a classification combining 4 perspectives: XAI Methods, XAI Evaluation, XAI perception, Human in the Loop [43]. We argue the importance of linking literature to its relevant definitions, clearly identifying the purpose of the research. This in turn will inform policy makers and government and will address some of the issues of interdisciplinary approaches to regulate AI we identified in the Introduction. We now highlight the definitions we will use throughout the rest of the document:

- Explanation: Within our document, our definition of explanation refers to some output from a system that is some combination of the above stages. We argue that an ideal artificial explanation must have direct contributions from stage 1-3 alongside some consideration of its evaluation (Stage 4) and an idea of where it will fit in deployment (Stage 5).
- Explainability: How well a system can produce an explanation of the above form
- Interpretability: A measure of how well a human understands a given explanation
- Transparency: Associated with Stage 1 of the explanation process, is a possible property of an explainable system.
- Justification: Associated with Stage 1 of the explanation process, are a type of post-hoc analysis made by a system to provide some possible causes for an output and is a possible property of an explainable system.

3.1 Stage 1: Identifying (Possible?) Causes

Phase 1 is concerned with methods in the XAI literature that attempt to collect the possible causes for a particular outcome. This phase of the explanation generation process has attracted the most attention in the literature and can be characterised as approaches that look at extracting causes from inherently transparent models and extracting rationalisations for outcomes already determined by the model (post-hoc explanations). While reading the following it is worth considering the argument made in the Introduction concerning the ongoing debate between causal reasoning and explanation. Many of the following techniques approach explainability from a machine learning perspective, neglecting the constraints of causal reasoning. In light of this, these techniques must not be seen a a complete solution to explainability - indeed many of these techniques have been recently associated with issues of robustness and manipulation [44].

3.1.1 Transparent Models

Rudin expresses concern over the contemporary approach of designing solutions to high stakes problems using black box methods, She argues that instead of trying to make these models explainable we should instead be focusing on applying already interpretable models to these solutions [44]. Arriteta et al. define these types of models as "transparent". A model is transparent if it can by itself be understood completely. They continue to define 3 further categories of transparency as simulatability, decomposability and algorithmic transparency [45]. These models include linear and logistic regression, decision trees, bayesian models, rule based learning, generative additive models, and K nearest neighbours [45]. Molnar discusses how linearity and monotonicity aid interpretability and how the correct balance must be struck between predictive accuracy and loss of interpretability when adding interactions between features [16].

Arrieta et al. claim that linear and logistic regression models meet all three characteristics of transparent models. However, note the difficulty in maintaining this transparency when applied to large datasets as well as the need to translate this transparency into some sort of human understandable explanation [45]. Mood discusses the potential pitfalls with extracting explanatory meaning from logistic regression models, specifically how unobserved heterogeneity affects the possibility of comparing co-efficients across multiple models [46]. Outside linear and logistic regression other transparent methods include using Boolean Logic to extract explainable rules [47].

Deep learning is famous throughout the literature for bringing high predictive accuracy to a multitude of data-based problems. Rudin claims however, that there is often no significant difference in accuracy between these and simpler, more interpretable algorithms [44]. Similarly, Ba et al. question the value added by increasing network depth [48]. Hagras et al. however, argue that "transparency rarely comes for free" [49]. As data complexity increases so does the trade-offs between accuracy and interpretability [49]. Kim argues that an effective explanation is not necessarily entirely transparent but it should aid the user's understanding of a particular result [50].

A significant contribution to explainability in the literature is concerned with techniques that map complex non-transparent models to more treansparent models. These include extracting abstractions via decision trees from trained neural network [51] or generalising linear classifiers to explain deep networks [52] or combining transparent clustering techniques with deep neural networks to understand the model architecture [53] .

3.1.2 Rationalisations (Post-hoc Explanations)

Often described as post-hoc analysis, this group of techniques focus on methods that are themselves not transparent but are designed to reason about model

behaviour after the model has acted. Arietta et al. argue that this form of explanation reflects the way in which humans explain things themselves [45]. Sheh et al. argue that these sort of explanations are in fact, rationalisations [54]. Lipton describes post-hoc analysis as methods including "natural language explanations, visualizations of learned representations or models and explanations by example" [55]. Arrieta et al. categorise post-hoc approaches as either model agnostic or ungeneralisable [45].

Ribeira et al. argue that model agnostic approaches address the limitations of transparent approaches, offering model flexibility that treats the underlying model as a black box, allowing for the choice of any machine learning model [56]. Arrieta et al. categorise model agnostic approaches as either model simplification, feature relevance explanations or visual explanation techniques [45].

Model simplification methods are the most common form of post-hoc explanation and are based on techniques for feature importance extraction, including local explanations. Examples of this form of explanation technique include LIME [57] which introduces local surrogate models as a method to approximate the conditions of the underlying black box model [16] or GREX, Genetic Rule Extraction [58], extended in [59] to show how the explanations generated by the method for rule extraction impacts accuracy.
Feature relevance explanation is described by Arrieta et al. as techniques that attempt to rank the relevance of each feature [45]. These techniques include, SHAP [60], unifying LIME and the game theoretic Shapley Values to explain individual predictions [16]. Explanation by example involves studying activation clusters [45].

Model specific approaches include post-hoc methods that attempt to explain shallow ML models that are not inherently transparent. This can include tree ensembles or random forests, made explainable by [61] who extracts variable importance from a random forest and support vector machines, made explainable by [62], who use an hyper-rectangle rules extraction to extract rules from a support Vector Machine.

The most popular methods for explanations of deep neural networks are local explanations and feature extractions [45]. Zhang et al. adapt convolutional neural networks to assign each filter layer with an object which generates explanations [63] whereas Wisdom et al. design interpretability for recurrent neural networks by modelling a a group of correlated obervations with a set of sparse linear vectors [64].

Dovsilovic et al. argue that for a human to trust a model approximation, we must understand over which domain it is reliable and accurate. This could be misleading for many pursuing local explanations [65]; coupled with the intractability of other model agnostic methods and the inability to generalise model specific methods, Dosilovic et al. argue we must turn to the social sciences to offer alternative approaches to explainability [65]. These sorts of approaches

are considered in Stage 2.

3.2 Stage 2: From Cause To Explanation

This stage of the explanations generation process is concerned with methods in the literature that begin to consider how best to generate human-interpretable explanations. Anjomshoae et al. observe that the majority of recent surveys consider only the explainability of data driven algorithms [38]. Similarly, Mittelstadt et al. argue that much of the recent explosion of activity in explainable AI is centred around one or two ideas within a much larger sphere of research. They continue to argue that the vast majority of this research works to provide approximations of complex models [15]. A key difference between research in Stage 1 and research in Stage 2 is the consideration of a human within the design process. Anjomshoae et al. argue that humans anthropomorphise agents to allow them to explain their behaviour. Similarly, Madumal argues that it easier for AI to adopt human-like explanations as these are easier for users to grasp [66]. Anjomshoae et al. argue that we must therefore turn to the social sciences to start explaining this behaviour [38].

Madumal suggests that causal models represent the method of explanation most closely reflecting that of human reasoning [66]. Mittelstadt et al. suggests that explanations are contrastive, social and selective to argue that study in this area should focus on contrastive, communicative explanations [15]. Contrastive explanations have been explored by comparing necessarily present and absent features for classification [67]. Walton combines contrastive explanations with argumentation suggesting that not only should explanations transfer knowledge of the causal chain but should also give argumentative support of this knowledge [68].

3.3 Stage 3 Communication of Explanations

Stage 3 of the explanation generation process is concerned with the literature that looks specifically at how best to communicate explanations.

Explanations can be delivered numerically, graphically/visually, in text and as apart of a formal argumentation framework. Numerical explanations are those considered in the Stage 1, outputting relative feature importance and weightings [57]. Graphical and visual explanations can include techniques such as saliency graphs for images [69] or partial dependence plots [70]. Other visual explanation techniques include a visual analytic interface that shows connections between instance level explanations [71], or a visual tool that uses decomposition to generate explanations of predictions that are then averaged and visualised to compare [72].

Computational argumentation involves reasoning with uncertainty by making use of logic to formalize the presentation of arguments and counterarguments

that are capable of dealing with conflicting information. There has been much work translating argumentation frameworks to generate explanations under various conditions [[73], [74]]. Recently there has been interest in using argumentation frameworks to address the interpretability of ensemble methods [75].

Madumal argues that understanding how humans engage in conversation is essential to generate meaningful explanations and suggests dialogical argumentation as a framework for this. Moulin et al. argue that from a theoretical perspective, argumentation and explanations are two indistinguishable constructs [76]. Building on Madumal's idea of communication, Walton [77] identifies that examining dialogical structures between two communicating parties would be valuable for the future progression of explainable agents but identifies the infancy of the study of this relationship. Fiedler et al. attempt to map machine reasoning to human adequate argumentation and show how this theoretic model can be applied to intelligent tutouring systems [78]. More recently, [79] present a dialogical argumentation based framework for extracting explanations for movie reviews [79].

Communication of explanations extends beyond a one-off output. Many explanations of systems need to be interactive and adapt over a relationship with a user. These can include interactive explanations of personal AI devices such as a smart home device or chatbot. Interactive explanations may incorporate user feedback into future explanations [80], adapt explanations to suit the user [81].

3.4 Stage 4 Evaluating Explanations

We have discussed at length the many open ended questions existing in the explainable AI domain. The principle challenge, however,is the question surrounding "what constitutes a successful explanation?" Evaluation of explainable AI is therefore a challenging area of research, again plagued by a lack of standard definitions, attracting much speculation but not yet producing tangible solutions.

In their survey, Giplin et al. [40] identify completeness and interpretability as the two main axes across which to gauge the success of an explanation. Explainability methods in the literature may try to evaluate their approaches via a) evaluating how well it explains the original proxy model [47] b) how well it approximates a secondary attribute of the original model e.g. saliency maps and model sensitivity [69] c) how well a method reveals the original model's bias [82] d) human evaluation [83] [40]. When creating explainable AI **for humans**, we argue that human evaluation should always be used for testing success. When evaluating explanations designed for humans further metrics should be considered including how explanations increase trust and satisfaction, Sokol identifies further evaluation metrics of these kind including coherence, soundness and contexfulness are ideal properties for human explanations [84].

3.5 Stage 5 Explanations in Practice

A key area that is often left out of existing surveys literature reviews despite being critical in the future of explainable AI is the process of taking explanations from a laboratory environment and applying them successfully in industry.

Alphabet's Google and International Business Machine Corporation have recently launched explainability toolboxes consisting of implementations of aforementioned theoretic research in feature importance and transparent models [60], [52], [47]. These toolboxes are intended to be applied to businesses who intend to deploy interpetable models using Google/IBM infrastructure yet these techniques are still in their infancy, each vulnerable to a particular set of trade-offs and in no way ready to be used in a regulatory capacity indeed it is not clear if these metrics are being adopted in any capacity at all across industry. This observation is validated in the survey conducted by Bhatt et al. who noted that the majority of organisations currently use explanations only as an internal debugging mechanism [85].

Health is one of the industries in dire need of explainable AI; the use of high stakes black box AI technology is ubiquitous across the sector. There has been a significant number of recent trials, implementing explainable AI systems within hospitals [86], [87]. However, there are still so many questions surrounding providing reliable explanations that must be answered before they can be used as regulatory options in high stakes scenarios.

4 Inter-disciplinary Background

4.1 Importance of Social Science Understanding

Tim Miller, through his 2017 survey, finds that the majority of current research in explainable AI does not develop existing theories from social science [88]. Mittelstadt et al. argue that this statistic derives from the wide and navigationally difficult gap between machine learning and the explanation sciences [15].

When exploring the semantic gap between the social sciences and computer science, Alan Cooper argues that when systems are designed by programmers they neglect the interactive component [89]. A collection of researchers in explainable AI have recently begun to argue that explainable AI is more than "simplified approximations" [65] of a model. Mittelstadt et al. argue that this research aims to recreate scientific explanations rather than cognitive explanations. Despite this, Mittelstadt et al. argue that these explanations are not technically scientific explantions as they only represent causal relationships between variables and do not necessarily have to apply by scientific laws [15]. Miller separates these approaches from the explanation sciences by building on Hilton's argument that causal explanation is a conversation [90], suggesting a difference between causal attribution and causal explanation [27].

Miller gives 3 major findings from the social sciences that are not currently widely studied in the literature but could offer insight to explainable AI [15]: contrastive explanations, selection of explanations, social process of explanations [27]. We discuss these ideas more closely in the following section.

4.2 Cognitive Process

Miller argues that an explanation is composed of a cognitive and a social process. The study of the cognitive process questions how humans perform causal connections, select and evaluate explanations [27].

4.2.1 Abductive Inference

Miller argues that as a cognitive process, explanations can be seen as a form of abductive reasoning. Abductive reasoning is one of three types of inference, the others being deductive and inductive. In contrast to the other two forms of reasoning, abduction is a non-necessary inference that takes into account explanatory and observed forces [91].
Pierce is considered the forefather of abductive reasoning [92]. Harman builds on Pierce to describe abductive reasoning as the "inference to the best explanation" and recent work in cognitive science and philosophy has supported Harman's claim [93]. However, Mcauliffe questions this link as abduction does not consider a hypothesis necessarily true [94]. Mcauliffe, however, agrees with Miller over the importance of studying the relationship between abduction and

different forms of inference to qualify scientific explanations, further philosophical research, understand consciousness and develop explainable AI [27].

Bridewell and Langley use an abductive inference framework to describe "everyday inference" that rationalises how humans use abduction to generate explanations [95]. Similarly, Khemlani links causal reasoning with abduction, arguing that a fundamental aspect of humanity is our ability to make explanations that rationalise our environment [96]. In contrast, [97] conduct a study to observe that humans display a preference for the most simple explanation. Khelami argues that humans actually prefer a causal explanation instead of a simple deductive explanation [96].

Rheder presents causal model theory that underpins the way in which humans assign probabilistic meaning to causal relationships [98]. Similarly to Khemlani, Rheder links causal mechanisms to abductive reasoning by expanding his original study to consider what properties of explanations allow observed generalisation in human behaviour [99]. Rheder notes that when similarity and causal explanations compete to generalise a feature, similarity is superceded by the strength of the causal explanation.

The study of abduction from a cognitive perspective has been applied to AI research. [100] presents an abductive reasoning framework for formulating diagnoses generalisable to recognition system and [101] uses an abductive reasoning framework in plan recognition. More recently, [102] use abductive inference in time series interpretation and [103] attempt to replicate human creativity by using abductive inference to play childhood story games. Abductive inference is even being linked to the future of creating artificial life [104]. Miller uses the link between generalisation, learning and abductive inference to argue the importance of further study within explainable AI, particularly to investigate how humans require less explanation as they learn more about a system.

4.2.2 Explanation Selection

We select explanations based on whether they are abnormal or unusual [105]. Hesslow expands this idea to emphasisze the difference between abnormal and unexpected, a subtlety hinging on the subjectivity of expectation. Hesslow provides a group of factors also influencing expectation selection including, perceived interest of the explainer, guilt and replaceablility. Miller discusses how the study of abnormalities can be used to study XAI; as humans they interact with increasing numbers of cases within the system they should become less surprised by abnormalities [27].

Prasada et al. [106] present evidence from empirical experiments that show how humans show a preference for universal factors (a dog has four legs) over individual differences (a dog is brown) when linking a type to its properties. Cimipian et al. develop this idea to suggest that humans rely on an "inherence

heuristic" that refers to the preference humans show for inherent features when selecting explanations. [107]. Miller argues how this human bias could be used to tailor explainable AI to mirror human behaviour [27].

4.2.3 Explanation Evaluation

Miller highlights the importance of considering the purpose of an explanation to guide its evaluation. Here, he argues that probabilities and likelihoods are not necessary criteria to optimise when evaluating an explanation [27]. If the overall goal of an agent is to select the most likely cause for an event, then a probability based explanation may be relevant. Howevever, if the overall goal of an agent is to build trust with a human then a more interpretable, user-friendly explanations may be preferable [27].

Leake proposes a goal based method of explanation evaluation proposing evaluation dimensions including, predictive power, distinctiveness and desirability to capture the variation of requirements in explanations [108]. In contrast [109] defines seven desired criteria of an explanation, including "effectiveness, transparency and trust". Tintarev argues that these dimensions may be conflicting and argues the importance of identifying the evaluation measure prior to explanation design. It is interesting that these evaluation metrics reflect the ambiguity in the definitions surrounding explainable AI. It is clear that if we are to evaluate and compare the literature, we must work towards some standard definition framework or at least attempt to contextualise individual works by identifying their place in the overall literature.

4.3 Social Attribution in Explanations

We have previously discussed how attribution theory plays an important role in the philosophy of explanation. Social attribution is the study of how humans explain the behaviour of others. For explainable AI, it is important for considering the expectations of explanations as well as the study of explanation selection and evaluation.

4.3.1 Perception and Folk Psychology

Heider defines social attribution as "person peception" [110] which captures the idea that social attribution assigns intent to people that may or may not be true. This concept is considered by folk psychology, the study of the way in which humans **think** they act. While the study of folk psychology is deeply embedded in psychology, Miller argues its relevance to the study in explainable AI [27]. De Graaf et al. argue that humans calibrate trust in intelligent autonomous agent by assigning psychological models of human behaviour to them. Drawing on folk psychology, De Graaf et al. expand this theory to argue that to engineer effective explainable agents, intelligent agents must be able to distinguish between intentional and unintentional behaviour [111]. Broekens et al.

integrate the idea of intentionality to their experiments, investigating the need for different explanations to describe different actions of intelligent agents [112].

Miller argues that many systems within AI are built upon folk psychology. Belief Desire Intention (BDI) agents have recently been applied to the application of virtual training for fire fighting staff using their inherent understanding of their own beliefs and desires to generate human-like explanations that in turn, aid training. [113].

4.3.2 Collective Behaviour

Miller argues that research into attribution of group behaviour is particularly relevant for those studying the areas of multi-agent planning, computational social choice and argumentation [27].

Susskind et al. discuss the differences between attribution and explanations of individual behaviour vs. group behaviour. They discuss the trend in the literature to study both of these separately and investigate whether this is due to inherent differences between the two. They argue that the cognitive processes are different due to the perceiver's expectations of behaviour rather than the inherent behaviour of the two types [114]. Intentionality is also studied within group attribution. O'Laughlin and Malle report that while perceivers expect less intentionality in multi-agent systems than individual agents, they in fact, expect more intentionality when looking at groups of agents acting together [115]. Recently, Ahlbrecht and Winikoff explored how explanations of agent behaviour could be used to understand the behaviour of collection of agents within a traffic simulation [116].

4.3.3 Norms and Morals

Lombrozo et al. investigate the use of social norms on attribution of behaviour. They claim that when evaluating underlying mental states, behaviour that conforms to social norms is less informative than behaviour that contradicts the norm. They claim that studying mental models is important in explanation generation yet should be studied in conjunction with normative consideration [117]. Miller argue that study of social norms is clearly valuable when developing anthropomorphic AI but argues its relevance to wider AI [27].

When engineering explainable AI, Miller argues that explanations that describe behaviour that contradicts social norms may be considered "bad" [27]. This subjectivity raises similar ethical questions we see across broader AI delivery and emphasizes the importance of not just blindly accepting explanations as correct, they should be subject to the same ethical analysis we are seeing applied across the industry. This idea is extended by Vamplew et al. who recognise the difficulty in creating AI that is beneficial to society as a whole. They develop a framework for balancing multiple potentially conflicting factors within

an explanation as a maximum expected utility function [118].

4.4 Explanation as a Social Process

Miller argues that explanation is part of a social interaction between an explainer and an explainee.

4.4.1 Explanation as conversation

Hilton defines a causal explanation as "first and foremost a conversation" [90]. He builds on counterfactual theory to develop a model of explanation that shows both how conversational explanations are influenced and constrained by interpersonal relations and how conversational explanations build on the social science perspective of cognition we have previously discussed [90]. Maduma et al. extend this idea to develop an interaction protocol that can explain the behaviour of AI systems interactively to users [119]. Similarly, Antaki et al. use argumentation to develop a framework for explanation conversation that involves arguing over potential and counter claims [120]. In contrast, Goyal et al. use counterfactual visual representations to to generate explanations and argue that their method results in higher user satisfaction [121]. Miller argues that all explanations should be conversational despite the media used [27].

5 Multi-Agent Systems

The study of multi-agent systems (MAS) works to establish mechanisms for the construction of complex systems involving multiple interacting individual agents [122]. Despite MAS being a relatively young area of research, it builds upon the contributions of Distributed AI (DAI) that has existed as a research field since the mid 1970s [123]. Distributed AI brings together several interdisciplinary concepts including computer science, sociology, mathematics, philosophy and is concerned with modelling intelligent agents pursuing some set of goals or performing some set of tasks [124]. DAI is comprised of three AI research areas: parallel AI, distributed problem solving and multi-agent systems [125]. Parallel AI deals with methodologies used to solve classical AI problems; distributed problem solving considers how a problem can be split over resources in computing entity and multi-agent systems focuses on how a problem can be solved across multiple computing entities or intelligent agents.

5.1 Intelligent Agents

Autonomy is the central idea behind an agent such that each agent will have a repetoire of actions available to it, will not have complete control of its environment but will be able to influence it. The definition of an agent, accepted across the literature is that of Russell and Norvig who define an agent as being able to perceive its environment through sensing and influence its environment through actuators [126]. Mathematically, an agent's behaviour is described by an agent function that maps an agent's perception to an action which is encapsulated by the agent program, a physical implementation that consists of a sequence of these actions [126].

Russel and Norvig categorise the different kinds of agent program as a simple reflex agent, where an agent selects action based on only the current perception of the world, a model based reflex agent, where the agent maintains some history of the world, a goal based agent, where as well as perception of the current world the agent has a pre-determined set of goals that it must act to obtain and utility based agents, which extend goal based agents to compare different actions and associated outcomes to determine how much of the goal had been achieved [126]. Russel and Norvig also define a rational agent as an agent that does the "right thing", depending on its environment and captured by a performance measure. An agent should be able to learn from experience and therefore gain autonomy - where what it learns from its perceptions outweigh what it knew a-priori [126].

Agent architectures are frameworks for decision-making that are embedded in an environment. Stuart defines several environment characteristics that determine the behaviour of an agent [127]. These include, fully observable vs partially observable environments where an agent can obtain up-to-date information about its environment's state; deterministic vs. stochastic environments where an action has a single, certain outcome; episodic vs sequential environments where

the behaviour of agent is dependent on a discrete episode and there is no link between episodes; static vs dynamic environments where the behaviour of agents is the only thing that changes the environment; known vs unknown environments where the laws of the environment are known by the agent [127].

Russell and Norvig define multi-agent systems as another form of agent environment whereas Belaji et al. define a multi-agent system as a group of loosely connected agents that interact by either their capacity for co-operation or their capacity for sharing information. Multi-agent systems can manifest in various different ways and therefore classification is complex. Stone and Veloso [122] note the complexity of multi-agent systems and categorise them using two axes of heterogeneity and level of communication. Parunak et al. present a taxonomy of MAS from an application perspective [128], whereas Stone et al. present a categorisation based on heterogeneous vs. homogeneous and communicative vs. non communicative agents [122]. More recently Dorri et al. have presented a categorisation under a wider variety of features including heterogeneity, decision function and leadership followed by a categorisation based on domain [129].

Stone and Veloso consider situations that are appropriate for multi-agent system solutions [122]. They argue that domains requiring MAS consist of multiple people or organizations with different goals and proprietary information. They continue to discuss some advantages of using multi-agents to deliver a solution. These include a speed up of system, associated benefits of paralellism, scalability and the simplicity of modularity [122]. Stone and Veloso also argue the potential utility of creating intelligent machines that mirror human behaviour by first creating social machines [122]. They point out the distinction between multi-agent systems and multi-robot systems, claiming that the latter can take advantage of "geographic distribution" [122].

5.2 Modern Age of Multi-Agent Systems

This next section illustrates how the state-of-the-art in MAS has changed over the past 20 years. In doing this, we argue how the problem of generating explanations for modern MAS is becoming increasingly pertinent.

Stone et al.'s taxonomy of MAS from 2000 classifies recent works by examining how they approach the theoretic predator/prey model, first introduced by Benda [130], applied to MAS to conceptualise how interacting agents work towards a common goal. The categories of MAS include homogeneous non communicating multi-agents, including deliberative and reactive agents, implemented in [[131],[132]]; heterogeneous non communicating multi-agents, implemented in [133]; homogeneous communicating multi-agents, implemented by [134]; heterogeneous communicating multi-agents.

At the turn of the century the volume of research was greatest in heterogeneous communicating multi-agents techniques which included Belief Desire Intention

(BDI) models and language models [122]. Similarly, in a 1998 survey, Iglesias discusses the recent trend of studying BDI models but noted the gap between theory and practical applications [135]. Similarly, in their study in 2000 Ricorde et al. noted the recent shift of MAS from research laboratories to software engineering [136].

Stone and Veloso give examples of domains in which to apply theoretic MAS advancements: design, planning, entertainment, games, air-traffic control, air combat, personal assistants, load-balancing, and robotic leg control [122]. These applications are removed from the general public and reflect the state of technology at the time. Lesser et al. in their 1995 article argue that "The next generation of applications alluded to will probably involve all the emphases of these generic applications and more" [137]. Over the next 20 years, AI agents became increasingly ubiquitous within society and MAS became increasingly complex. Today, we can see multi-agent systems in user facing applications such as autonomous vehicles, intelligent tutouring systems, hospital scheduling, ride sharing, scheduling algorithms and recommender systems. Further, we can see MAS research occurring across a wide variety of domains including, cloud computing where [138] designs an agent based approach to co-ordinating bag-of-things applciations in the cloud; social networking where Gatti et al. use a stochastic multi-agent based approach to predict user behaviour on Twitter [139]; security where Sakira et al. present a MAS approach to prevent users from tabnabbing (stealing private informationby impersonating reputable sites) [140]; routing, where Claes et al. present a MAS approach to anticipatory vehicle routing [141]; robotics, where Rosa et al. design a co-operative control scheme for unmanned aerial vehicles [142]; modelling complex systems, where Domiguez et al. present an agent based model approach to modelling supply chains [143]; agents in cities and environments where Van et al. present a multi-agent market approach to control heating in buildings [**?**].

We can also see the explosion of work within multi-agent systems reflected in established global communities. Several conference series devoted to the study of agents began in the 1990s, including the International Workshop on Agent Theories, Architectures, and Languages (ATAL), the International Conference on Autonomous Agents (AGENTS), and the International Conference on Multi-Agent Systems (ICMAS). In 2002, these three merged to form the International Joint Conference on Autonomous Agents and Multi-Agent Systems (AAMAS). From 2002 to 2019 the topics within AAMAS have become increasingly diverse with submitted papers ranging from re-inforcement learning, agent societies and economic paradigms.

An increasing number of conferences centred on intelligent agents have been founded in the last two decades. In 1998 the International Conference on Principles and Practice of Multi-Agent Systems (PRIMA) was founded and in 2002 the conference for Practical Applications of Agents and Multi-Agent Systems (PAAMS) was founded. 2009 saw the introduction of the International Confer-

ence on Agents and Artificial Intelligence (ICAART). 2019 has seen the introduction of the first "International workshop on Explainable Transparent Agent and Multi-Agent Systems" (EXTRAAMAS). EXTRAAMAs was started to address four central issues within the community:

"(i) To establish a common ground for the study and development of explainable and understandable autonomous agents, robots and MAS. (ii) To investigate the potential of agent-based systems in the development of personalized user-aware explainable AI. (iii) To assess the impact of transparent and explained solutions on the user/agents behaviors. (iv) To discuss motivating examples and concrete applications in which the lack of explainability leads to problems, which would be resolved by explainability." [144]

5.3 State-of-the-art Explainability for Multi-Agent Systems

This section motivates the study of explainability for multi-agents systems and compares current approaches to those described in Chapter 2, showing how recent works in single explanation generation could be adapted to a MAS setting.

When considering multi-agent systems from an explainability perspective we can categorise systems as either human and agent systems or multiple agent systems. Multiple agent systems include swarm technology, simulations, financial agents and smart grid technology. The explanatory complexity in these kinds of systems arises from bounded resources available. A resource could be the level of responsibility of the system, time available to achieve a goal or level of trust in individual agents [145]. In time critical situations explanations must be tailored for delivery efficiency. In trust critical situations, explanations should be tailored to increase trust between agents. How these explanations propagate through the system, when they are delivered in respect to actions and how they are aggregated to explain the overall system are critical in future study. In [144], Calverisi et al.look at ways in which blockchain technology can be utilised to augment user trust in real world MAS applications and Ahlbrecht et al. investigate using explanations to generate knowledge about multiple agents in simulations [116].

Ensemble methods are a further example of multi-agent systems. Ensemble methods are used to combine base models into an ensemble that often results in higher predictive accuracy [75]. [75] propose a novel approach that addresses the lack of interpretability in ensemble methods by developing a framework that combines outputs from Deep Neural Networks using multi-agent argumentation.

Human, robot systems are described by EXTRAAMAS as systems whereby succeeding in collaboration necessitates a mutual understanding of the status of other agents/users/ their capacities and limitations [144]. Richardson et al. note the widespread emergence of these kind of systems in domains such

as “digital personal assistants, recommendation systems, training and tutoring systems, service robots, natural language processing or chat bots, planning and self-driving cars” [6]. We also note the emergence of human-agent teams in domains such as disaster recovery, space exploration and firefighting. One key difference between explanations required for these systems rather than the kinds of system considered in Chapter 2 is the quality of information shared between system components (whether they be human or agent) during interactions. This perspective motivates our exploration of explanation as a conversation, discussed in Chapter 3. Bourgais et al. consider the interaction between several agents, presenting an agent architecture for explaining the behaviour of social agents [146]. The review conducted by Richardson et al. considers this interaction from a single perspective: that of the internal explanations required of the data-based components of the system. This, we equate to the discussion of state-of-the-art in Chapter 2. We argue that the added explanatory complexity of a multi-agent system originates from the interactions between agents. This complexity can manifest in many ways.

Consider a system comprised of multiple individual components that all output (potentially conflicting) decisions and associated explanations. Imagine a medical diagnoses system that recommends a particular drug to treat a patient. The system is comprised of multiple AI agents analysing medical scans, an AI Echocardiography agent that detects irregular heartbeat patterns and a non-adherence algorithm that determines a patient’s risk of not taking their prescribed medicine. The computational challenge in this scenario is the aggregation of the decisions, how the explanations could be used to inform this aggregation and a further aggregation of the explanations themselves. A further challenge in this scenario is how the aggregated decision and explanation is outputted to the medical professional in order for them to sustain sufficient trust to implement the overall decision. Sassoon et al. address this problem by applying explainable argumentation to the clinical consultation domain [147] and Jentzsch et al. propose conversational interfaces as a way to present explanations to the user [148].

The medical professional could either be considered part of the multi-agent system or separate as some kind of implementer. We argue that considering them as part of the system allows for a better explanation, in line with ”truly explainable AI” as expressed by Doran et al. [18]. We therefore argue that for any human facing intelligent agent, it is preferential from an explanatory perspective to view this as a multi-agent system, even if the system is comprised of solely a data driven algorithm and an end user.

Considering a multi-agent system from a human-in-the-loop perspective adds a further level of complexity to the system. This presents itself as the way in which an explanation is presented to the user. Richardson et al. discuss the importance of using explanations to generate trust in the system [6]. Consider a system that involves an automated algebra tutor and a child. To learn algebra,

the child will need to be supplied explanations as to why their attempt is wrong or correct. However, to effectively learn, the child may want to question an explanation further, ultimately requiring explanations of explanations. We argue that considering the human as part of the system will allow for the necessary formalism for this kind of interaction. Recent work that considers the interaction between humans and machines include [149], who present an explanatory piece examining the way in which transparency effects the behaviour of human and agent teams and [150], who look at the effect of explanations in human-robot interactions.

A new level of complexity is added when we consider multi-agent systems that have individual components operating with private or proprietary data. Consider a local forensic unit who use multiple algorithms including image recognition to detect child abuse. This type of approach, whilst proving effective, comes with a range of ethical and privacy issues. We argue that explanations could be used to address some of these issues. Firstly, explanations could be used to detect algorithmic bias. Secondly, from a systems perspective, explanations could be used to ensure private data does not leak across multiple components in a system. If component A needs output from component B where component B analyses protected data, an explanation could be tailored to output the necessary information needed by A, without exposing private data.

It is clear that much of the work in this space is still in its infancy. This is demonstrated by a) lack of consideration in ICO guidelines [14] b) lack of multi-agent content in EXTRAAMAS c) noted by kraus et al. [1], who in their survey provide mostly state-of-the-artin explainable machine learning chapter, describing this work as "basic ingredients" for explainable multi-agent systems.

6 Possible Future Directions

6.1 Argumentation as a mechanism to control Explanations in multi-agent Systems

Chapter 5 presents the motivation for studying the interaction between individual components of a multi-agent system and how this impacts explanation. Moulin et al. argue that explanation and argumentation are both used to control interactions within a multi-agent setting [151].

One prominent research area within argumentation is the challenge of selecting the best arguments and counter-arguments from an argument tree. Hunter proposes that selection can be aided by taking into account intrinsic and extrinsic factors of arguments [152]. We hypothesise that the application of cognitive processes discussed in Chapter 5, specifically the use of intrinsic and extrinsic properties, can be applied to this problem. From an explanation perspective this would be beneficial in evaluation, either to evaluate the aggregated explanation of the system or to evaluate the individual explanations within the system.

Sassoon et al. [147] apply argumentation to a clinical wellness consultation scenario to explore how argumentation could allow a conversation between patient and AI consultant.

6.2 User Modelling for Increased Satisfaction

A further area in which argumentation could be applied is in user modelling for increased explanation satisfaction. Chapter 2 discussed the myriad of approaches currently being used to generate explanations for AI. Alan Cooper argues that approaches to explainability designed by decision-making system experts do not consider the effectiveness of the explanation on the user [89]. One way of measuring the effectiveness of explanations is through qualitative studies as performed by . However, an alternative approach is to use argumentation user modelling that adapts an argumentation framework to its particular environment. This line of reasoning could also be applied to address the challenge of different users requiring different explanations from the same system or even to study how explanations should change depending on length of the relationship with a user-facing system. An alternative approach to user modelling is presented by Lahav et al. who use re-inforcement learning to learn what is interpretable to users [153].

6.3 Interactive Explanations

Motivated in chapter 5, this research area would look at how explanations could themselves be explained. This could be approached from an argumentation perspective, analysing how the argument tree could be traversed to provide deeper explanations.

6.4 Time dependent explanations

A further area under scrutiny in the review, motivated in Chapter 5, is the need for epxlanations within time constrained multi-agent systems. Temporal Argumentation Frameworks, as proposed by Budan et al. present an interesting framework within which to generate these explanations [154]. How this framework could be adapted to real world applications would be useful in generating quality explanations for time critical systems.

6.5 Secure Explanations

A further area for research highlighted by this review is the need for secure explanations. Calverisi et al. propose a framework that applies blockchain technology to explanations to ensure their security [144]. Future work in this area could develop frameworks that ensure the security of individual components while maintaining interpretability. A different research area for consideration within the remit of secure explanations, motivated in chapter 5, is how an argumentation framework could be adapted to generate explanations that themselves offer a secure transference of knowledge between system components.

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